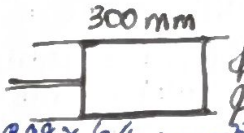
- FOR SOLIDS WE GENERALLY TALK ABOUT STRESS-STRAIN, BUT FLUIDS DEFORM CONTINUOUSLY WHEN SHEARED ⇒ MOTIVATION FOR TALKING ABOUT STRAIN RATES.
- SOME MATERIALS EXHIBIT COMBINATION OF SOLID + FLUID BEHAVIOR ⇒ VISCOELASTIC.
- RULES OF CALCULUS ARE APPLICABLE FOR (CONTINUOUS + DIFFERENTIABLE) FUNCTIONS → HELPS EASE ANALYSIS AND THAT IS MAJOR ROLE OF CONTINUUM ASSUMPTION!) → MENTION NOTATION (x, y, z, t)
- GIVE THE SYMBOLS FOR DENSITY (ρ), SPECIFIC GRAVITY ($SG = \rho/\rho_{H_2O}$), SPECIFIC WEIGHT ($\gamma = \rho g$)
- WHEN TALKING ABOUT VELOCITY, TALK ABOUT MASS-AVERAGED VELOCITY $\bar{v} = \Sigma m_i v_i / \Sigma m_i$, VOLUME AVERAGED VELOCITY, $v^{(k)} = \Sigma v_i^{(k)} / n$ } RELEVANT FOR MULTI-COMPONENT MIXTURE
- STATE OF STRESS CAN BE DESCRIBED COMPLETELY BY SPECIFYING THE STRESS ON ANY THREE MUTUALLY ORTHOGONAL PLANES → EMPHASIZE THE SIGN CONVENTION.
- FROM A GEOMETRIC CONSTRUCTION, WE SHOULD SHOW THAT $d\gamma/dt = du/dy$: $\delta w . \delta t = \delta l = \delta y . \delta \alpha$ ⇒ FOR A DIFFERENTIAL ELEMENT, WE GET.

- FROM DYNAMIC VISCOSITY, MOTIVAE THE CONCEPT OF KINEMATIC VISCOSITY.
- DESCRIBE SURFACE TENSION & BULK MODULUS OF A FLUID. $\{E_v = dP/(d\rho/\rho)\}$

EXAMPLE: A LUBRICANT HAS $\nu = 2.8\times10^{-5}\ m^2/s$, $SG = 0.92$. MEAN VELOCITY OF PISTON IS 6 M/S, DETERMINE POWER DISSIPATED.



APPROXIMATELY CARTESIAN, $\delta y = 0.1$ mm ⇒ $\tau = \mu\, du/dy = \nu\rho \cdot du/dy = 2.8\times10^{-5} \times 0.92 \times 998 \times 6/(0.1\times10^{-3})$

⇒ DRAG $= \pi \times (150\times10^{-3}) \times 0.3 \times \tau = 217.596$ N ⇒ $P = F \times v = 217.596 \times 6 \approx 1308$ W.

- FLUID STATICS HAS NO RELATIVELY VELOCITY ⇒ ONLY NORMAL FORCES ACT ⇒ $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$ → RELATED TO PRESSURE.
- WHY ARE WE STARTING WITH STATICS: NO FLOW, NO VISCOSITY, NO TURBULENCE
- MENTION HERE PASCAL'S LAW WITH A TRIANGULAR ELEMENT.
- BASIC EQN. OF FLUID STATICS, WE ASSUME CONSTANT DENSITY, { EXPLAIN SIGN CONVENTION RELATED TO PRESSURE

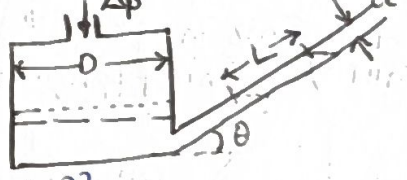
$dF_{BODY} = \rho \cdot g \cdot dx\,dy\,dz$, $dF_{SUR} = -(\partial p/\partial x\,\hat{i} + \partial p/\partial y\,\hat{j} + \partial p/\partial z\,\hat{k})\,dx\,dy\,dz = -\nabla p \cdot dV$ ⇒ $dF_{TOT} = (-\nabla p + \rho g)\,dV$

SINCE WE ARE DOING STATICS, NET FORCE IS ZERO ⇒ $-\nabla p + \rho g = 0$

(PRESSURE FORCE PER UNIT VOLUME IS BALANCED BY THE BODY FORCE PER UNIT VOLUME AT EVERY POINT)

- ABSOLUTE PRESSURE, RELATIVE PRESSURE, GAUGE PRESSORE { MINATE THAT THERE IS A DATUM }

EXAMPLE: AN INCLINED TUBE MANOMETER IS CONSTRUCTED. DERIVE A GENERAL EXPRESSION FOR THE LIQUID DEFLECTION L IN THE INCLINED TUBE, DUE TO THE APPLIED PRESSURE Δp. OBTAIN AN EXPRESSION FOR THE MANOMETER SENSITIVITY AND DISCUSS THE IMPACT OF D, d, θ AND SG.

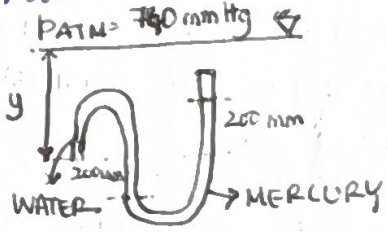


CONSERVATION OF MASS @ CONSTANT DENSITY ⇒ $\pi D^2 \times h_1 = \pi d^2 \times L$ —i)

PRESSURE VARIATION, $\cancel{p_{atm}} + \Delta p = \rho g h_1 + \rho g \cdot L \sin\theta + \cancel{p_{atm}}$ ⇒ $\Delta p = \rho g \cdot L\{\sin\theta + (d/D)^2\}$

SENSITIVITY = CHANGE IN OUTPUT/CHANGE IN INPUT $= \frac{1}{\rho g}\{\sin\theta + (d/D)^2\}$ ⇒ ρg ↑ THEN SENSITIVITY ↓, d/D ↑ THEN SENSITIVITY ↓ θ ↑ ⇒ $\sin\theta$ ↑ ⇒ SENSITIVITY ↓

EXAMPLE: CONSIDER THE FOLLOWING SUBMARINE SUBMERGED IN WATER. DETERMINE THE DEPTH OF THE SUBMARINE BELOW WATER (GIVEN $\gamma = 10^4$, 760 mm Hg = 101.3 kPa)



STATICS BASIC EQUATION ⇒ $p_{ATM} + \rho g y + \rho g \times (200\text{ mm}) - \rho_{HG} \cdot g \cdot 400\text{ mm} = p_{OBS.}$

GIVEN: p_{ATM} = 740 MM HG, $p_{OBS.}$ = 840 MM HG

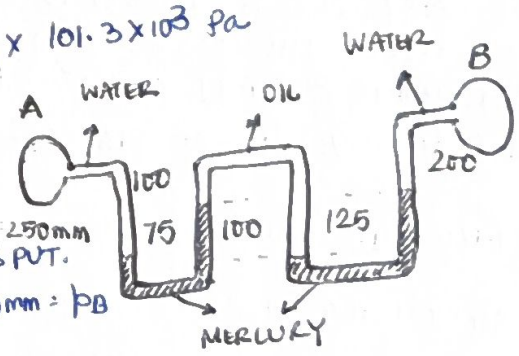
⇒ 100 mm HG $= \rho \cdot g \cdot y + \rho g\,(200\text{mm}) - 400$ mm HG

⇒ 500 mm HG $= \rho g \cdot [y + 200\text{ mm}]$

WE KNOW THAT, 760 mm Hg = 101.3×10^3 Pa (FACT) ⇒ 500 mm HG $= 50/76 \times 101.3\times10^3$ Pa

⇒ $50/76 \times 101.3\times10^3 = 10000\{y + 0.2\}$ ⇒ $y = 6.46$ m.

EXAMPLE: WATER FLOWS THROUGH PIPES A AND B. LUBRICATING OIL IS IN THE UPPER PORTION OF THE INVERTED U. MERCURY IS IN THE BOTTOM OF THE MANOMETER BENDS. DETERMINE $p_A - p_B$ IN KPa. (SG Hg = 13.6, OIL = 0.88)



ASSUMPTION IS THAT THERE ARE NO VELOCITY GRADIENTS WHERE MANOMETER IS PUT.

⇒ $p_A + \rho_{H_2O} \cdot g \cdot 250\text{mm} - \rho_{HG} \cdot g \cdot 75\text{mm} + \rho_{OIL} \cdot g \cdot 100\text{mm} - \rho_{Hg} \cdot g \cdot 125\text{mm} + \rho_{H_2O} \cdot g \cdot 200\text{mm} = p_B$

⇒ $p_A - p_B = g\{200\text{mm} \cdot \rho_{Hg} - \rho_{OIL} \cdot 100\text{mm} - \rho_{H_2O} \cdot 50\text{mm}\}$

- FORCE ON BODY, $F_R = \int p\,dA$. $\int (p_o + \rho g y \sin\theta)\,dA = p_o A + \rho g \sin\theta \int y\,dA$, BY DEFINITION $\int y\,dA = y_c A$ (CENTROID)

⇒ $F_R = p_o A + \rho g \sin\theta \cdot y_c A = (p_o + \rho g y_c \sin\theta)A = (p_o + \rho g h_c)A = p_c A$ (PRESSURE AT CENTROID)

- MOMENT OF RESULTING FORCE, $y' F_R = \int (p_o + \rho g y \sin\theta) \cdot dA \cdot y = \int p_o y\,dA + \int \rho g y^2 \sin\theta\,dA = p_o y_c A + \rho g \sin\theta \int y^2 dA$

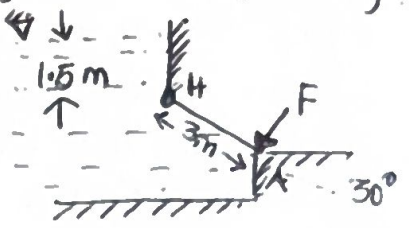
FOR A UNIFIED FORMULA, $I_{xx} = I_{\hat{x}\hat{x}} + A y_c^2$, WHERE $I_{\hat{x}\hat{x}}$ = MOMENT OF AREA ABOUT CENTROID

⇒ $y' F_R = p_o y_c A + \rho g \sin\theta [I_{\hat{x}\hat{x}} + A y_c^2] = A y_c (p_o + \rho g h_c) + \rho g \sin\theta\, I_{\hat{x}\hat{x}} = y_c F_R + \rho g \sin\theta\, I_{\hat{x}\hat{x}}$

- $y' > y_c$, WHICH MEANS THAT LOCATION OF THE RESULTANT FORCE IS ALWAYS BELOW THE LEVEL OF THE PLATE CENTROID FOR NORMAL CONFIGURATIONS.
- SIMILARLY, $x' F_R = \int (p_o + \rho g y \sin\theta) \cdot dA \cdot x = p_o \int x\,dA + \rho g \sin\theta \int xy\,dA = p_o x_c A + \rho g \sin\theta \cdot I_{xy}$

AGAIN FOR UNIFIED, $I_{xy} = I_{\hat{x}\hat{y}} + A x_c y_c \Rightarrow x'F_R = x_c(p_o + \rho g h_c) + \rho g \sin\theta\, I_{\hat{x}\hat{y}} = x_c F_R + \rho g \sin\theta\, I_{\hat{x}\hat{y}}$

EXAMPLE: THE GATE SHOWN IS HINGED AT POINT H. THE GATE IS 3m WIDE NORMAL TO THE PLANE OF THE DIAGRAM. CALCULATE THE FORCE REQUIRED @ A TO KEEP IT CLOSED



FLUID WILL EXERT A NET FORCE & MOMENT, THAT THE FORCE MUST BALANCE.

MOMENT BALANCE, $\int_{y=0}^{3m} (p_o + \rho g \times 1.5m + \rho g \cdot y \sin 30^\circ)\, y\, dA = F_x \times 3m$ (CANCEL BECAUSE OTHER AND ALSO ATMOSPHERE)

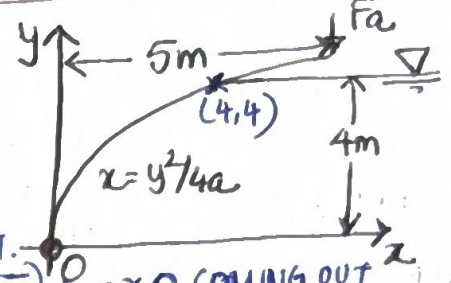
$\Rightarrow p_o H \cdot y^2/2\big|_0^3 + \rho g \times 1.5 \times H \cdot y^2/2\big|_0^3 + \rho g \sin 30^\circ \cdot H\, y^3/3\big|_0^3 = F_x \times 3m$

$\Rightarrow \rho g H \{1.5 \times 3^2/2 + \sin 30^\circ \times 3^3/3\} = F_x \times 3 \Rightarrow 1000 \times 9.81 \times 3m/3m \times \{1.5 \times 4.5 + \tfrac{1}{2} \times 9\} = 110.36\,kN.$

- $F_R = -\int p\, dA \Rightarrow F_R = F_{Rx}\hat{i} + F_{Ry}\hat{j} + F_{Rz}\hat{k} = -\int_{A_x} p\, dA_x\, \hat{i} - \int_{A_y} p\, dA_y\, \hat{j} - \int_{A_z} p\, dA_z\, \hat{k}$ (DOT PRODUCT WITH THE COMPONENT)

- RATIONALE FOR VERTICAL FORCE FORMULATION IS THAT PRESSURE DOES NOT VARY ALONG THE VERTICAL DIRECTION FOR COMMON ENGINEERING APPLICATIONS. ALSO, IF SURFACE HAS FLUID ON ALL SIDES THEN IT WILL BE IN EQUIBRIUM, WHICH MEANS IT IS SUPPORTING WEIGHT OF THE FLUID ABOVE.

EXAMPLE: THE GATE SHOWN IS HINGED AT O AND HAS CONSTANT WIDTH, $w = 5m$. THE EQN. OF THE SURFACE IS $x = y^2/a$, WHERE $a = 4m$. THE DEPTH OF THE WATER TO THE RIGHT OF THE GATE IS $D = 4m$. FIND THE MAGNITUDE OF THE FORCE F_a APPLIED AS SHOWN REQUIRED TO MAINTAIN THE GATE IN EQM. ASSUME THAT THE WEIGHT OF THE GATE IS NEGLIGIBLE.



THE FLUID EXERTS A HORIZONTAL FORCE, A VERTICAL FORCE WHICH WE SHOULD GET.

HORIZONTAL MOMENT = $\int_0^4 \rho g (4-y)\, dy \cdot W$ ⊙ (COMING OUT) (4−y) FROM DEPTH; FROM MOMENT

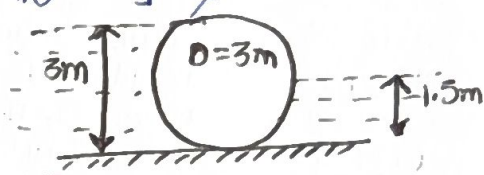
VERTICAL MOMENT $\int_0^4 \rho \cdot W \cdot (4 - \sqrt{ax})\, dx\, g$ ⊗ COMING OUT

EXTERNAL MOMENT = $5m \times F_a$ ⊗, EQM. $\Rightarrow \rho g W \int y(4-y)\, dy + \rho g W \int_0^4 x(4-\sqrt{ax})\, dx = 5 \cdot F_a$

$\Rightarrow \rho g W [4/2 \cdot y^2\big|_0^4 - 1/3\, y^3\big|_0^4 + 2x^2\big|_0^4 - \sqrt{a} \times 2/5 \cdot x^{5/2}\big|_0^4] = 5 \cdot F_a$

$\Rightarrow 999 \times 9.81 \times 5 [2 \times 16 - 1/3 \times 64 + 2 \times 16 - 4/5 \times 32] = 5 \cdot F_a = 167.256\,kN$

EXAMPLE: CONSIDER THE CYLINDRICAL WEIR OF DIAMETER 3m AND LENGTH 6m. IF THE FLUID ON THE LEFT HAS A SPECIFIC GRAVITY OF 1.6, AND ON THE RIGHT A SPECIFIC GRAVITY OF 0.8, FIND THE MAGNITUDE & DIRECTION OF THE RESULTANT FORCE.

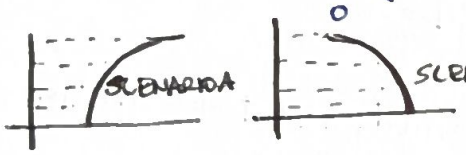


FLUID ON LEFT $\Rightarrow$ $3m \times 6m$ = AREA $\Rightarrow$ FORCE = $\int_0^3 (p_{ATM} + \rho_1 g h) \cdot 6\, dh$ → BALANCED BY ATM. ON FIRST QUADRANT

FLUID ON RIGHT $\Rightarrow$ $\int_0^{1.5} (p_{ATM} + \rho_2 g h) \cdot 6\, dh$ ← $\Rightarrow$ DIFFERENCE = $6 \cdot p_{ATM} \times 3 + \rho_1 g \cdot 6 \cdot h^2/2\big|_0^3 = \rho_1 g \cdot 6 \cdot 9/2$

$- 6 \cdot p_{ATM} \cdot x 1.5 - \rho_2 g \cdot 6 \cdot h^2/2\big|_0^{1.5} - \rho_2 g \cdot 6 \cdot 1.5^2/2$

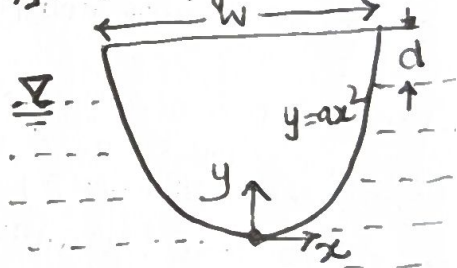
$= \rho_w g \cdot 6 [1.6 \times 4.5 - 0.8 \times 2.25/2]$



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- TOP SURFACE $h_1(x)$, BOTTOM SURFACE $h_2(x) \Rightarrow$ FORCE ON SURFACES = $\rho g [h_1(x) - h_2(x)] \cdot dA = \rho g\, dV$ ARCHIMEDES PRINCIPLE

EXAMPLE: THE CROSS-SECTIONAL SHAPE OF A CANOE IS MODELED BY THE CURVE $y = ax^2$, WHERE $a = 3.89$ AND THE COORDINATES ARE IN FEET. ASSUME THAT THE WIDTH OF THE CANOE IS CONSTANT $W = 0.6m$ OVER ITS ENTIRE LENGTH OF 5.25m. SETUP A GENERAL ALGEBRAIC EXPRESSION RELATING THE TOTAL MASS OF THE CANOE AND ITS CONTENTS TO DISTANCE d BETWEEN THE WATER SURFACE AND TOP OF THE CANOE. CALCULATE TOTAL MASS WITHOUT SWAMPING THE CANOE.



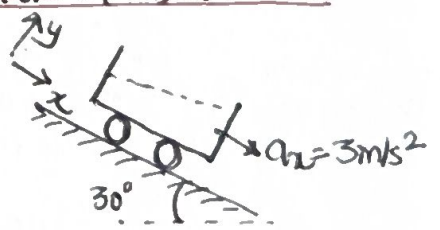
COORDINATE @ TOP OF CANOE = $(W/2, aw^2/4)$, WHERE IT MEETS WATER = $\left(\sqrt{w^2/4 - d/a},\ aw^2/4 - d\right)$

DISPLACED VOLUME OF WATER = $2\int_{x=0}^{\sqrt{w^2/4 - d/a}} \left(aw^2/4 - d - ax^2\right) dx \cdot L = 2L\left[\left(\frac{aw^2}{4} - d\right)\sqrt{\frac{w^2}{4} - \frac{d}{a}} - \frac{a}{3}\left(\frac{w^2}{4} - \frac{d}{a}\right)^{3/2}\right]$

SINCE THIS BALANCES THE WEIGHT OF CANOE $\Rightarrow Mg = \rho g \cdot 2L\left[\left(\frac{aw^2}{4} - d\right)\sqrt{\frac{w^2}{4} - \frac{d}{a}} - \frac{a}{3}\left(\frac{w^2}{4} - \frac{d}{a}\right)^{3/2}\right]$

- WHEN TALKING ABOUT STABILITY, NEED TO TALK ABOUT CENTER OF GRAVITY AND CENTER OF BUOYANCY.

EXAMPLE: A RECTANGULAR CONTAINER OF WATER UNDERGOES CONSTANT ACCELERATION DOWN AN INCLINE. DETERMINE THE SLOPE OF THE FREE SURFACE.



ALONG THESE COORDINATES, $g_y = -g\cos 30^\circ$, $g_x = g \sin 30^\circ$

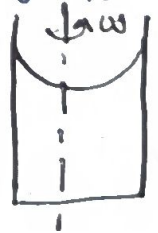
$\Rightarrow$ EQN FOR FLUID STATICS IS, $-\partial p/\partial x + \rho g \sin 30^\circ = \rho \cdot a_x$, $\frac{-\partial p}{\partial y} - \rho g \cos 30^\circ = 0$

$\Rightarrow \partial p/\partial x = (\rho g/2 - \rho a_x)$, $\partial p/\partial y = -\rho g \cdot \sqrt{3}/2$

ALONG FREE SURFACE, PRESSURE IS CONSTANT $\Rightarrow dp = 0 \Rightarrow \partial p/\partial x \cdot dx + \partial p/\partial y \cdot dy = 0$

$\Rightarrow (\rho g/2 - \rho a_x) \cdot dx - \rho g \sqrt{3}/2 \cdot dy = 0 \Rightarrow dy/dx = \frac{g/2 - a_x}{g \cdot \sqrt{3}/2}$

EXAMPLE: A CYLINDRICAL CONTAINER IS ROTATED @ ω. AFTER SOMETIME, THIS ROTATES AS A RIGID BODY. DETERMINE THE SHAPE OF THE FREE SURFACE.



$\nabla p = \partial p/\partial r\, \hat{e}_r + 1/r\, \partial p/\partial\theta\, \hat{e}_\theta + \partial p/\partial z\, \hat{k} \Rightarrow -\partial p/\partial x = -\rho \omega^2 x$, $-\partial p/\partial z - \rho g = 0$

$dp = 0 \Rightarrow \rho \omega^2 x \cdot dx - \rho g\, dz = 0$ | INTEGRATE B/W SUITABLE LIMITS.